

Visual Comparative Analysis of Process Variants

Interview questions

A. Demographic information

- In which country do you currently work?
- What sector do you work in?
- What is your role with respect to process mining in your organization?

B. Experience in process mining

- How many years of experience do you have in the field of process mining?
- How many years of experience do you have in the field of process mining?
- Which process mining tools do you use most frequently?
- How many process mining projects related to process analysis have you been involved in? For how many different organizations?

C. Process Mining Analysis.

- When you first start working with an event log, do you typically start with a specific hypothesis, such as that one process is performing better than another, or do you take a more exploratory approach to discover insights, variations, or anomalies?
- If you are conducting an exploratory analysis, what are the main steps you typically carry out during this phase?
- Do you usually use process visualizations such as BPMN diagrams or DFGs (Directly-Follows Graphs)? If so, how do they help in your analysis?
- Do you ever compare different versions of the same process? If so, what kinds of insights are you usually looking for, and how do you typically carry out this comparison?
- When comparing models, do you usually focus on just two at a time, or you have to deal with comparing more than two models at once?
- Do you find DFGs useful in these comparisons? Why or why not?

- What aspects or information shown in DFGs do you consider more relevant to compare them?
 - Some tools let you compare process models using side-by-side views or layered views. Do you use any features like these? Have you seen other ways to compare different versions of the same process in existing tools?
 - Which challenges or limitations have you faced when trying to compare process models using existing tools?
 - Do you miss specific features in existing tools to better support model comparison?
 - Would it help if similarities and differences were automatically highlighted using color coding or annotations?
 - Is there anything else you'd like to add about how you compare processes or use tools in this area that we haven't covered?
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Workshop

A. Tasks performed with the tool

The first phase of the evaluation involves performing a series of tasks using the tool.

- Scenario 1: Identify structural differences between organizational unit 65482' and the rest of organizational units.
- Scenario 2: Identify similarities in the process fragment `` from Start trip to Declaration REJECTED by EMPLOYEE" across the cases associated with each resource (Staff member vs. System).
- Scenario 3: Identify missing activities between the cases associated with the Employee and Budget Owner roles, excluding any activities related to the permit process.

B. Technology Acceptance Model

The second phase encompasses the assessment of the perceived usefulness and perceived ease of use of the tool by the subjects, as measured by the Technology Acceptance Model. Responses were on a 7-point Likert scale ranging from 1 (strongly disagree) to 7 (strongly agree).

B.1. Perceived usefulness

- I would find the tool useful for visual process comparison.
- Using the tool would increase my productivity in visual process comparisons
- Using the tool would enhance my effectiveness in visual process comparisons.
- Using the tool would make it easier to carry out visual process comparisons than with the existing methods
- Learning to operate the tool would be easy for me
- My interaction with the tool would be clear and understandable.

B.2. Perceived ease of use

- It would be easy for me to become skillful at using the tool.
- I would find the tool easy to use.
- I would find it easy to get the tool to do what I want it to do

- I would find the tool to be flexible to interact with.
- The visual comparison helped me spot relevant differences quickly
- I understood what each visualization represented.

C. Qualitative feedback questions

- What did you like about the comparative features?
 - How would this scale to your real datasets?
 - Which visual encodings or comparison strategies are missing?
 - What would you like to improve about the comparative features?
 - Other comments
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